

## Assignment 8

CSC250

Due Date: November 13, 11:59pm. Points: 50

Single PDF file named: A8\_\*.pdf where \* is your last name.

### Question One: (5 points)

- 1. If a computer system allows eight-character passwords comprising a random combination of decimal digits and letters A-Z (uppercase or lowercase), what is the maximum number of passwords that a user will have to try with the brute force method of breaking into a user's account? Why?**

If a computer system allows an eight character password with a random combination of decimal digits (0-9) and letters (A-Z) - (a-z), it would be around 218 trillion password attempts to break into an account using the brute force method. Because there are 62 possibilities per character in the eight character password ( $10(0-9) + 26(A-Z) + 26(a-z) = 62$ ), it is raised to the power of eight to include the other characters in the password ( $62^8 = 218,340,105,584,896$ ).

- 2. Suppose that a hacker is trying to guess a password in the above system. Further, suppose that the system forces a five-second delay after each password guess. How long will it take the hacker to guess the password in the worst-case analysis? Why? Show all your work.**

To find how long it would take for the hacker to break into the account, the following formula should be followed since the number of passwords is known already - (Time = Passwords \* TimeDelay). Then, (Passwords \* TimeDelay/Time(Days) =  $62^8 * 5/86,400 \approx 12,645,290,192$  days using worst case analysis to find how long it would take to find the password with a 5

second delay. The value 86,400 comes from multiplying the amount of seconds in a minute (60) with the amount of minutes in an hour (60) to the amount of hours in a day (24), and then finally multiplying the # of password attempts originally by 5 to include the five second delay.

**Question Two: (5 points)**

**1- What are the purpose of the set-user-ID (SUID), set-group\_ID(SGID), and sticky bits?**

The set user ID is a bit used to differentiate between a superuser and any other user when executing a file it helps with giving the appropriate privileges to the appropriate users.

SGID allows permissions of the owner of the file's group. So, the owner can give access to a group of other users that only have access to that one file. Then, the sticky bit, if it is on, means that any user can create a file or directory in a certain directory; however, they cannot delete a file or directory they don't own.

**2- How many bits are used by LINUX to store a file's type and permissions? Show the arrangement of these bits.**

16 bits are used to store a file's type and permissions. And, can be shown by the following:

File type (4 bits)

User permissions (3 bits)

Group permissions (3 bits)

Owner permissions (Sticky bit - (3 bits))

**Question Three: The user Bell sets access permission to his home directory by using the command `chmod 700 $HOME`. If the file `hw4.cpp` in his home directory has read permission to 777, can anyone read this file? why or why not? (10 points)**

If `hw4.cpp` in the home directory has file permissions 777, then anyone can read, write, and execute the file. The reason is because each digit represents an octal value, and if each bit is '7' then every bit for that octal value is '1'. Each octal value represents the permissions for each group, (user, groups, other). So, anyone can read/write/execute the `hw4` file but only the user has full permissions over the directory (700).

**Question Four: Create a file, called `sample`, in your home directory and set its permissions to read and write for yourself, read for users in your group, and none to everyone else. (5 points)**

**1- What command did you use to create the file?**

```
[o.manley94@quincy.edu@thor ~]$ touch sample
```

```
-rw-r--r-- 1 o.manley94@quincy.edu domain users@quincy.edu 0 Nov 13 19:45 sample
```

**2- What command did you use to set privileges?**

```
[o.manley94@quincy.edu@thor ~]$ chmod 640 ~/sample
```

```
-rw-r----- 1 o.manley94@quincy.edu domain users@quincy.edu 0 Nov 13 19:45 sample
```

**Question Five: Calculate the file access permissions assigned to newly created directories, executable files, and text files for bit mask 027. Show all your work. (10 points)**

Directory: bit mask = 027.

000010111

$C' = 111101000$

$B = 111111111$

$A = B \text{ and } C' = 000101000 = 050$  (octal representation for directory calculation).

Executable file: bit mask = 027.

000010111

$C' = 111101000$

$B = 111111111$

$A = 111101000 = 750$  (octal representation for executable calculation).

Text file: bit mask = 027.

000010111

$C' = 1111010000$

$B = 101101101$

$A = 101101000 = 550$  (octal representation for text file).

**Question Six : Suppose that you have a text file, called SecureData, that contains important company data and you have set 600 permissions on it. Further, suppose that you have an executable program called SecureUpdate that allows you to update contents of text files, including SecureData. You set permissions on SecureUpdate to 511 and set its SUID bit. Do you see any problems? Explain your answer. (15 points)**

Because SecureUpdate has its SUID bit turned on, anyone who accesses this program will have the privileges of the file owner. Which could lead to issues where unsolicited users will be given access to info they shouldn't be having as well as the possibility of tampering with other files.